

Agroverse — Robotics-for-Cacao Barter

Goal for barter trade: use **machine maintenance cost** as the pricing anchor for cacao beans — bypassing the volatile commodity price of cacao.

§0 · The goal in one line

Farmers get robotic help (pruning, harvesting, roast control) — and pay with **cacao valued against machine maintenance cost**, not against the volatile commodity price. Both sides get a **stable, predictable unit of exchange**; farmers keep cash, Agroverse gets beans, and nobody is at the mercy of the commodity ticker.

Why this matters (current market): NY cocoa swung **\$6,065 (Dec 2025) → \$3,157 (Apr 2026) → ~\$4,205 (May 2026)** — roughly a 48% swing in five months. South American prices fell from **\$9.11/kg (Q1 2025) to \$7.50/kg (Q1 2026)**, **–18%** while other regions fell far more. A barter priced in *commodity cacao* would force both farmer and robot-partner to gamble on a coin flip. **Machine maintenance cost is the stable anchor** — an engineering cost that does not swing with the weather, disease, or futures market.

§1 · The three problems

Problem 1 — Witches' broom. Canopy disease caused by *Moniliophthora perniciosa*. Infected brooms must be **cut off and collected** — **physically removed from the plantation** (not left on the ground, not sprayed). Manual detection + pruning is slow, skilled, and easy to miss. Losses commonly reach **30–50% of yield**. **No chemical spray: this is an agroforestry system, so witches' broom is managed by mechanical removal, not treatment.**

Problem 2 — Aging farmer population & labor shortage. Farm owners are aging; the next generation is not stepping in. Farm management and harvest depend on increasingly scarce, expensive manual labor. Know-how is concentrated in aging owners — and when the work gets physically too hard or someone gets hurt, the farm gets sold.

Problem 3 — Roast consistency. Roasting is a sequential decision process with **delayed reward** — you act on the roast curve, and only learn after the batch whether the flavor profile landed. Today it is skilled craft, tuned batch-by-batch by experienced tasters: hard to reproduce, hard to scale.

Problem 4 — Farm succession risk. When a farmer is hurt or **too old** and the farm is sold, there is **no guarantee the new owner continues agroforestry** (they may clear, replant, or run it down). Every farm lost = regenerated hectares **permanently reversed** — a direct threat to the **10,000-hectare regeneration goal**. The agroforestry system lives or dies with the individual owner — that is fragile.

§2 · Working prototype reference — Frasky (IIT)

Video: <https://youtu.be/hg8qYrjyYCU>

Built by the **Italian Institute of Technology (IIT)** with Bergamo-region agricultural stakeholders (roboticist **Dr. Manuel Catalano**). Frasky autonomously:

- **Monitors** grape clusters (camera + digital twin of every cluster's location)
- **Manipulates** plants (robotic arm)
- **Applies targeted spray treatments**

One deliberate difference — no spray: Frasky sprays; **Agroverse does not**. We borrow the *mechanical* capability (monitor → manipulate → act) — **not** the chemical one. In agroforestry, witches' broom is managed by **cut & collect**, never by spraying.

Frasky (grape)	Cacao adaptation	Problem	Feasibility
Cluster monitoring via camera	Broom/pod detection via vision	P1	■ tractable now
Plant manipulation (robotic arm)	Cut & collect infected brooms (mechanical, no chemicals)	P1	■ tractable now
Targeted spray	Not adopted — agroforestry, no chemical inputs	—	■ deliberately excluded
—	Selective pod harvest (varied heights, trunks + branches)	P2	■■ harder R&D
—	Farm-management data (monitoring, records for aging owners)	P2/P4	■ tractable now

§3 · Second application — roast control as a reinforcement learning problem

The reframe: roasting is a **sequential decision process with delayed reward** — the agent acts on the roast curve, and only learns *after* the batch whether the flavor profile landed. That is exactly **reinforcement learning (RL)**, not a static recipe.

RL element	In the cacao roaster
State (observation)	Bean moisture, temp / rate-of-rise, roast color, particle size, volatiles
Action	Heat, airflow, drum speed, batch timing
Reward	Flavor-profile match + consistency + energy efficiency
Policy	Learned roast curve reproducing a target profile, batch after batch

- **Each batch = one training episode**; reward arrives at the end; the controller adapts to bean lot, moisture, machine state to hit the flavor target

- **Particle detection = the observation layer** of the RL loop (the agent is only as good as what it can sense)

Detection approach	Measures	Status
Machine vision	Bean color, browning, particle size	■ proven in food lines
Hyperspectral / SWIR imaging	Moisture, roast degree, defects (slaty/moldy)	■ proven, conveyor-scalable
Portable NIR spectroscopy	Chemical roast markers, in-line	■ proven, low-cost
Electronic nose + ANN	Volatile profile → roast degree (94.4% reported)	■ proven

Precedent: IMA Group's "AI Learning to Roast" + model-based roast optimization — real, but expensive industrial software. Our pitch: a **low-cost RL roast controller tuned to the Agroverse network's beans** → turns flavor control from craft into a **scalable, measurable process**.

§4 · The proposal — barter: robotics for cacao

One robotics/AI partner (design expert), four workstreams:

Problem	Robotic/AI response	Feasibility
P1 · Witches' broom	Broom detection + cut & collect robot (mechanical removal, no chemicals — agroforestry-aligned)	■ tractable now
P2 · Aging farmers / labor	Harvest assist + farm-management data	■ tractable (harvest ■■ R&D)
P3 · Roast consistency	RL roast controller + particle detection	■ parallel, lower-field-risk
P4 · Succession risk	Automation extends ownership longevity (below)	■ structural, low-tech-risk

How the barter works:

1. Partner provides robots + maintenance (design expert)
2. Agroverse provides farmer network, market access, logistics
3. Farmers provide **cacao as payment**
4. **The anchor: cacao is valued against machine maintenance cost** — a stable engineering cost — **not the volatile commodity price**
5. Agroverse takes the commodity-market side; farmers and partner are insulated from price swings

Why P4 matters most: P1–P3 improve the farm; **P4 keeps the farm from being lost**.

§5 · The causal chain — automation → ownership longevity

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Automation takes over the heavy, high-injury work

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Less physical strain on the aging farmer — fewer injuries, less exhaustion

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The farmer can keep running the farm safely and productively, years longer

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Fewer forced sales from injury or "too old to work" — ownership longevity extended

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Farms stay in agroforestry hands → succession risk drops

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Regenerated hectares stay standing → 10,000-hectare goal holds

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The root cause of the forced sale is often **injury or labor that is simply too hard for an older body**.

Automation directly attacks that root cause. It is not just "less work" — it is **more years of safe, productive ownership**.

P4 structural protections:

1. Farm runs with far less physical labor → owner stays productive longer
2. Farm-management data makes the operation **ownable/transferable** (know-how isn't trapped in the owner's head)
3. The cacao-barter agreement carries an **agroforestry-continuity clause** → value stays tied to the standing trees

The system's goal: make an agroforestry cacao farm *more valuable as agroforestry than as cleared land* — the strongest protection for regenerated hectares.

§6 · Pilot plan (1 season, gated)

1. **Gary chats with farmers about receptiveness to using robotics** — gate 1. The first gate, before any robotics commitment. Result determines everything downstream.
2. **Once farmers are receptive → invite the partner to visit the farms to check the terrain** — gate 2. Site visit so robot design is grounded in real conditions: tree spacing, canopy height, slope, wet-season mud, row access.
3. **Scoping call with the robotics partner** — share the Frasky video (youtu.be/hg8qYrjyYCU) as the reference prototype; MVP framing: broom pruning/monitoring tractable now, selective harvest is R&D; evaluate the roast-RL track (state sensors, reward signal, data capture per batch).
4. **Formal farmer pitch (PT-BR)** — labor shortage + witches' broom yield-loss + aging-owner relief framing; no cash outlay; paid in cacao; no chemicals (agroforestry-compatible).

5. **2–3 test farms** — route through the Bahia operations (Ilhéus) + an existing farm network.
6. **Barter accounting** — value cacao at machine-maintenance cost, record into the supply ledger at fair market value.
7. **Measure, then scale** — % brooms removed, harvest kg, labor-hours saved, farmer-reported strain/injury reduction, years of continued ownership (delayed succession), batch-to-batch roast consistency.

§7 · Next steps & owners

#	Step	Owner
1	Chat with farmers about receptiveness to robotics (gate 1)	Gary
2	When receptive → invite partner to visit farms to check terrain (gate 2)	Gary
3	Scoping call with robotics partner (Frasky video + MVP + roast-RL track)	Gary + partner
4	Approve outreach copy / draft question list	—
5	Formal farmer pitch (PT-BR)	Agroverse — Bahia operations (Ilhéus)
6	Select 2–3 test farms	Agroverse
7	Barter accounting (machine-maintenance-cost anchor)	Supply ledger

§8 · Mission tie-back

Agroforestry cacao is the engine of regeneration. Every problem above — broom, aging farmers, roast inconsistency, succession — is a reason a farm might fail, get sold, or get cleared. Robotics + AI, paid in cacao, keeps the engine running:

- Every broom **cut and collected** keeps the canopy alive
- Every aging farmer given a **tool instead of a shovel** keeps ownership in agroforestry hands
- Every roast made **consistent and scalable** builds the market that pays for the trees
- **Automation that keeps the farmer safely owning and farming longer is automation that keeps agroforestry land out of the clearing market**

Continuing agroforestry becomes the profitable choice — that is how we reach and hold 10,000 hectares.

Draft v13 · pending approval before any outreach.